Resampling

Letting the computer do what the formulas were invented to avoid

Permutation tests · Bootstrap · Sample size from a pilot study

The plan for this session

1 · Permutation test

Is the difference between two groups more than chance?

2 · Bootstrap

How precise is my estimate — a median, a ratio, anything?

3 · Pilot studies

How many patients does the real study need?

- Two ideas do all the work: reshuffle the labels, or redraw the patients.
- Every example is base R on plain vectors — no packages, no data frames, no models.
- You will be able to run all of it on your own data by the end of the afternoon.

HISTORY

Four tests before the computer, one after

1900

Chi-square test

Karl Pearson

mathematician, London

Counts in a table vs. counts expected if nothing is going on.

1908

t-test

W. S. Gosset – "Student"

brewer at Guinness, Dublin

Small samples of barley and beer.

1935

Fisher's exact

R. A. Fisher

statistician, Rothamsted

Small 2 × 2 tables where chi-square fails.

1945

Wilcoxon

Frank Wilcoxon

chemist at American Cyanamid

Ranks instead of values: quick by hand, no normality. Bug sprays

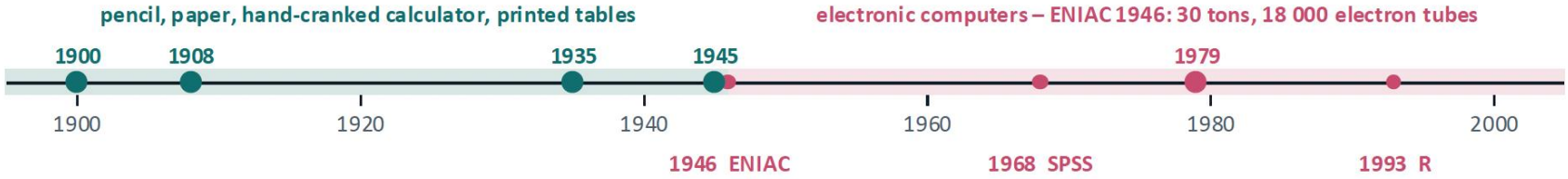
1979

Bootstrap

Bradley Efron

statistician, Stanford

Re-draw your own sample thousands of times: a CI for anything, no formula. Permutation.



The classical tests were built for pencil and paper – hence critical-value tables, and ranks instead of values. Resampling is the one idea that had to wait for the machine: thousands of re-draws give a p-value or a CI for anything – a median, a ratio, clustered teeth – with no formula and no normality.

THE PROBLEM

Where the classical tests come from

- Student's t-test: 1908. Fisher's ANOVA: 1920s. No computers, anywhere.
- **Every calculation was done by hand, and the answer had to come out of a printed table.**
- **So a test assumes a shape for the data, derives a formula from that shape, and looks the answer up.**
- Those assumptions are artefacts of hand computation — not claims about biology.

Assume

↓

Derive

↓

Look up

ASSUMPTION 1 OF 3

The statistic is t-distributed — not the data

the patients — cost, skewed, and we know it

the mean of 30 of them — near enough t

What it assumes

Mean follows a t distribution (normalish distribution adjusted fo sample size)

Why that matters

Does the mean of your sample behave normal?

ASSUMPTION 2 OF 3

The central limit theorem has not finished working

n = 5 — skewness still 0.89

n = 30 — skewness down to 0.36

ASSUMPTION 3 OF 3

A standard error formula exists for your statistic

mean

$$SE = s / \sqrt{n}$$

in every textbook

mode

?

in none of them

What it assumes

That someone has already done the algebra. True for means, proportions, order statistics

Where it genuinely fails

The mode. A ratio of two medians. Cost per millimetre of pocket reduction. The 90th percentile of waiting time. No textbook covers these.

THE PROBLEM

What a p-value actually asks

The question behind every test

If the treatment did nothing at all, how often would chance alone hand me a difference as big as the one I am looking at?

- **Everything else — t, F, chi-square — is just machinery for answering that one question.**
- **The classical route answers it with algebra derived from an assumed curve.**
- **Resampling answers the very same question by simulating 'the treatment did nothing' directly.**

The idea

Stop assuming a distribution. Make thousands of copies of your own study instead, and look at how much the answer moves.

What varies across those copies is exactly the uncertainty you were trying to quantify.

Reshuffle and Bootstrap

Reshuffle the labels

Same measurements, new group assignments. Every value used exactly once.

→ permutation test: is there a difference?

Redraw the patients

Draw a new sample from your own data, with replacement. Values may repeat.

→ bootstrap: how precise is the estimate?

ADVANTAGE 1 OF 5

No shape is assumed

- The method uses your distribution, not an idealised one.
- Skewed, bounded, lumpy, ordinal — it does not care.
- Nothing has to be transformed to make the algebra behave.

ADVANTAGE 2 OF 5

Statistics without a formula worth trusting

- Median
- Difference between two medians
- The ratio of two medians
- Cost per millimetre of pocket reduction
- The 90th percentile of waiting time

The rule

If you can compute it once from a sample, you can resample it.

ADVANTAGE 3 OF 5

It suits small samples

- **Classical tests lean on large-sample approximations.**
- **Those approximations are weakest exactly where clinical research lives: 8, 12, 20 patients.**
- **A permutation test makes no such approximation — with 12 patients it can enumerate every possible outcome.**

ADVANTAGE 4 OF 5

You can see it happen

- **The null distribution appears as a histogram on the screen.**
- It is no longer a number in a table
- Far easier to explain to a co-author, a reviewer — or a student.

ADVANTAGE 5 OF 5

It costs no extra patients — but it is not magic

What it gives you

More out of the data you already have: honest intervals, exact p-values

What it cannot do

Invent information. Repair selection bias or confounding. Rescue a study that was badly designed.

Which one answers which question

YOUR QUESTION	METHOD	WHAT YOU REPORT
Is the difference more than chance?	`Permutation test`	a p-value
How precise is my estimate?	`Bootstrap`	a 95% confidence interval
How big must the real study be?	`Bootstrap a pilot`	a sample size, and its range

1

The permutation test

Shuffle the labels, and see what chance alone can produce.

The question, and the data

Two treatments for periodontal pockets. Twelve patients, six in each arm. Outcome: probing depth after treatment, in millimetres.

Treatment A	3.1	2.8	4.0	3.6	2.9	3.4	3.30 mm
							Difference 0.78 mm
Treatment B	2.4	2.1	3.0	2.6	2.2	2.8	2.52 mm

Six per group is small — and entirely typical of a clinical pilot. A t-test here would be leaning hard on its assumptions.

PERMUTATION TEST

The null hypothesis

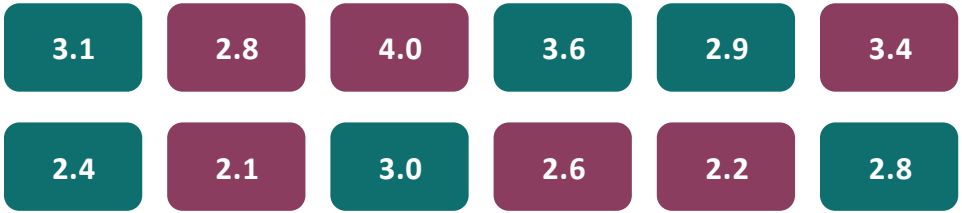
If the treatment did nothing, then each patient's depth is simply their depth — they come from the same distribution**. So forget the labels.**

3.1	2.8	4.0	3.6	2.9	3.4
2.4	2.1	3.0	2.6	2.2	2.8

In R this is one line: pool <- c(a, b)

PERMUTATION TEST

Deal them out again, at random



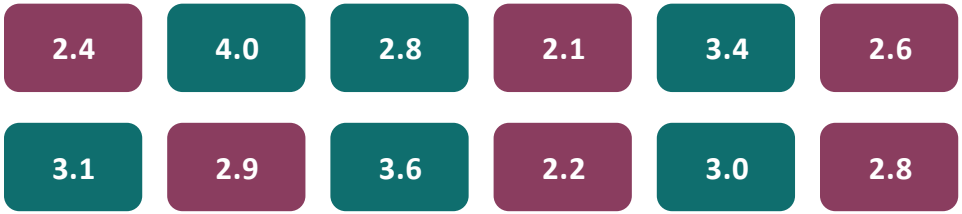

```
new A: 3.1, 3.6, 2.9, 2.4, 3.0, 2.8   →   mean 2.97
new B: 2.8, 4.0, 3.4, 2.1, 2.6, 2.2   →   mean 2.85
```

Same twelve numbers. Only the colours moved — exactly what the null hypothesis claims happens.

0.12

difference from this deal

And again. And again.

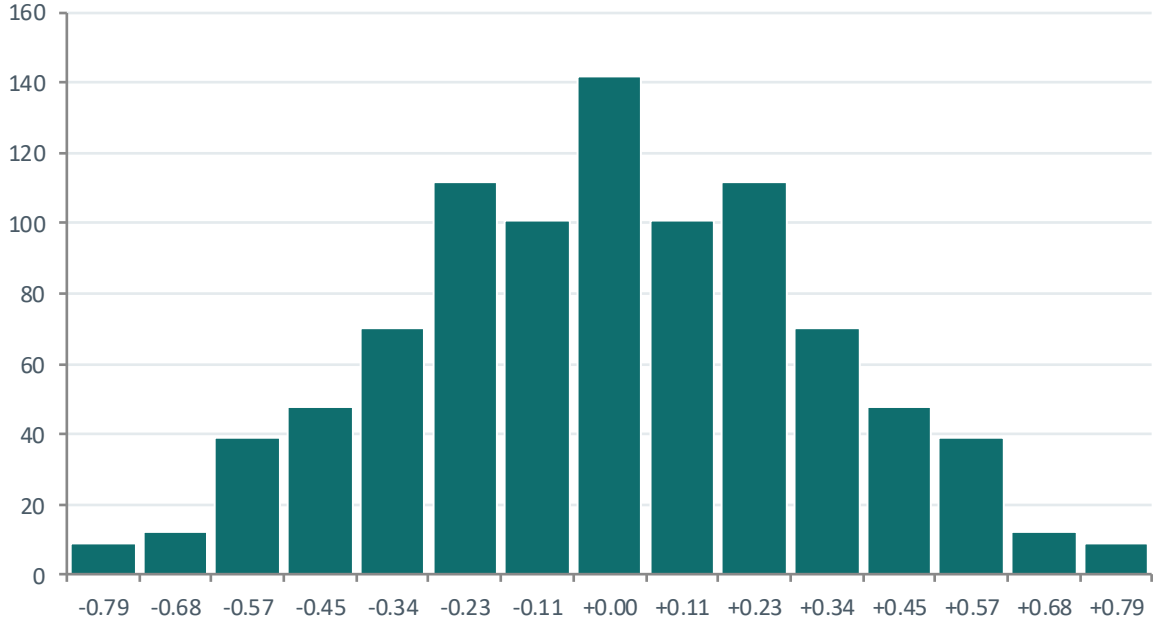


deal 1	+0.12
deal 2	−0.35
deal 3	+0.41
deal 4	−0.08
…	…

Collect them all

Each deal gives one difference that chance alone produced. Thousands of deals give the whole picture.

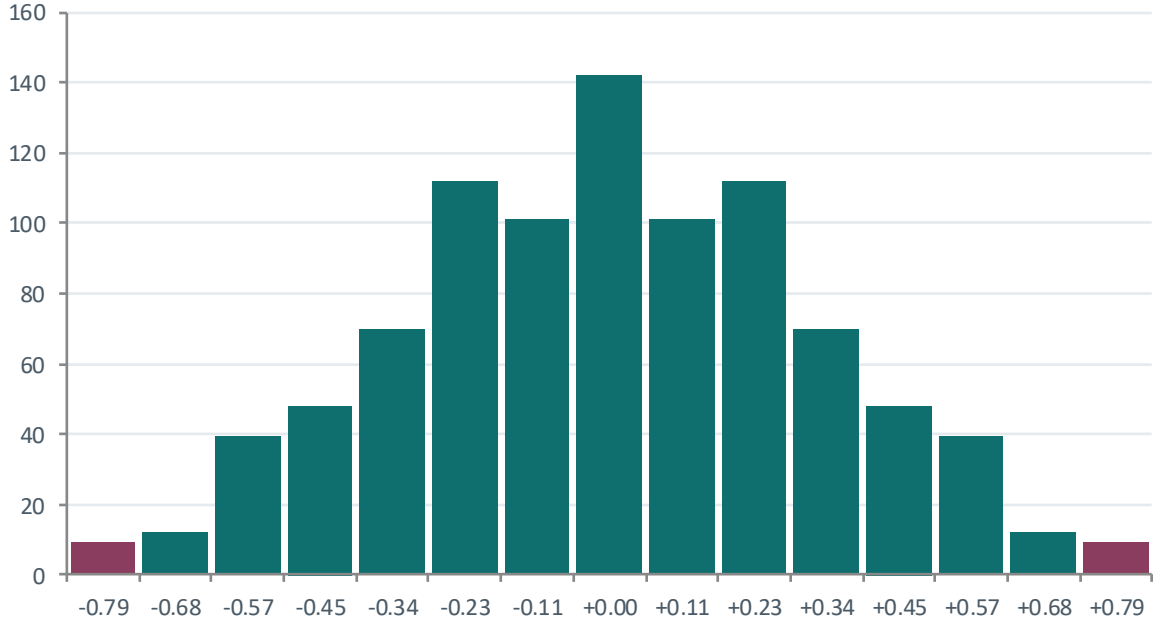
Every possible deal



Difference in means produced by chance alone (mm)

This is the null hypothesis

Where does 0.78 fall (sample difference)?



0.011

of all deals were this extreme

`p = 0.011`

Coloured bars: deals at least as extreme as the one we observed

The whole thing, in R

```
a <- c(3.1, 2.8, 4.0, 3.6, 2.9, 3.4)
b <- c(2.4, 2.1, 3.0, 2.6, 2.2, 2.8)

obs  <- mean(a) - mean(b)
pool <- c(a, b)

diffs <- replicate(10000, {
  i <- sample(12, 6)                # deal the labels at random
  mean(pool[i]) - mean(pool[-i])
})

mean(abs(diffs) >= abs(obs))        # the p-value
hist(diffs); abline(v = obs, col = "red")
```

Line by line: the observed difference

```
obs  <- mean(a) - mean(b)      # 0.78 — the number we must judge
pool <- c(a, b)                # all twelve values in one vector
```

- obs is computed once. It never changes again.
- pool stacks the two groups into a single vector of twelve numbers — the twelve cards, face up.
- After this line, nothing in the data records which treatment a value came from.

Line by line: sample(12, 6)

```
i <- sample(12, 6)      # six positions out of twelve, no repeats
```

- Draws six numbers from 1…12, without replacement.
- It picks positions, not values.
- One call might give 3 7 1 12 5 8; the next gives something else.

Without replacement

Every card is dealt exactly once. Contrast this with the bootstrap later, where cards are put back.

Line by line: pool[i] and pool[-i]

```
pool[i]      # the values at those positions   ->  the new group A
pool[-i]     # minus = "all the others"        ->  the new group B
```

The minus sign

In R, a negative index means leave these out. It is not a negative element and not a subtraction.

replicate() and the p-value

```
diffs <- replicate(10000, { ... })     # 10 000 deals, all kept

mean(abs(diffs) >= abs(obs))           # what share was as extreme?
```

- replicate runs the block over and over and collects the answers into one vector.
- abs() on both sides makes the test two-sided: extreme in either direction counts.
- mean() of a TRUE/FALSE vector is a proportion — that proportion is the p-value.

How does it compare with a t-test?

p = 0.011

permutation test

p = 0.007

two-sample t-test on the same data

Same conclusion, different footing

The t-test assumed a normal distribution to get there. The permutation test assumed only that the labels were interchangeable. With six per group, that is a much cheaper assumption.

2

The bootstrap

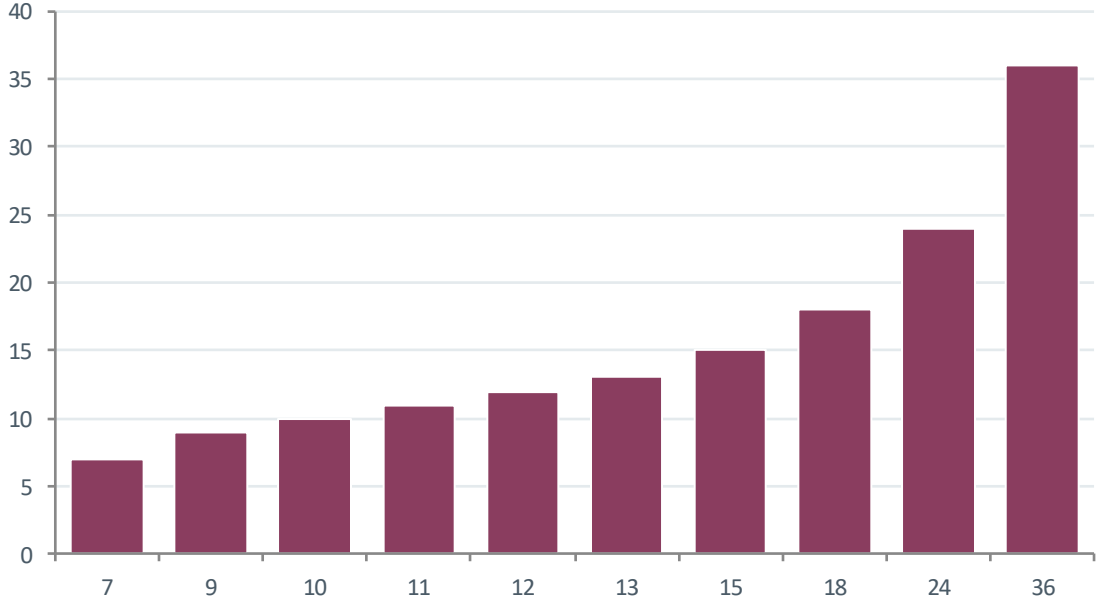
Redraw your own patients, and watch how far the answer moves.

A different question

A 95% CI without a formula, without an idealized distribution

- For a mean, the textbook hands you a formula for the confidence interval.
- **For a median, a ratio, a 90th percentile — there is no formula.**
- Yet those are often the quantities that matter clinically.

The data: waiting time for an appointment



Ten patients, days waited, sorted

12.5

median wait, days

And the interval?

Ten patients is not many. How far could that 12.5 be from the truth?

Why there is no formula for a median

- **The interval for a mean rests on the central limit theorem — means of samples behave predictably.**
- **Medians do not. Their behaviour depends on the shape of the distribution near the middle.**
- **Which is precisely the thing you do not know.**

The way out

BOOTTRAP THE MEDIAN

The bootstrap idea

The leap

Your sample is the best picture of the population you will ever have. So treat it as the population, and draw new studies from it.

- Draw ten patients from your ten, **with replacement** — some appear twice, some not at all.
- Compute the median of that new sample. Write it down.
- Do it 10 000 times, and look at the spread of the 10 000 medians.

One bootstrap sample, by hand

12.5 days

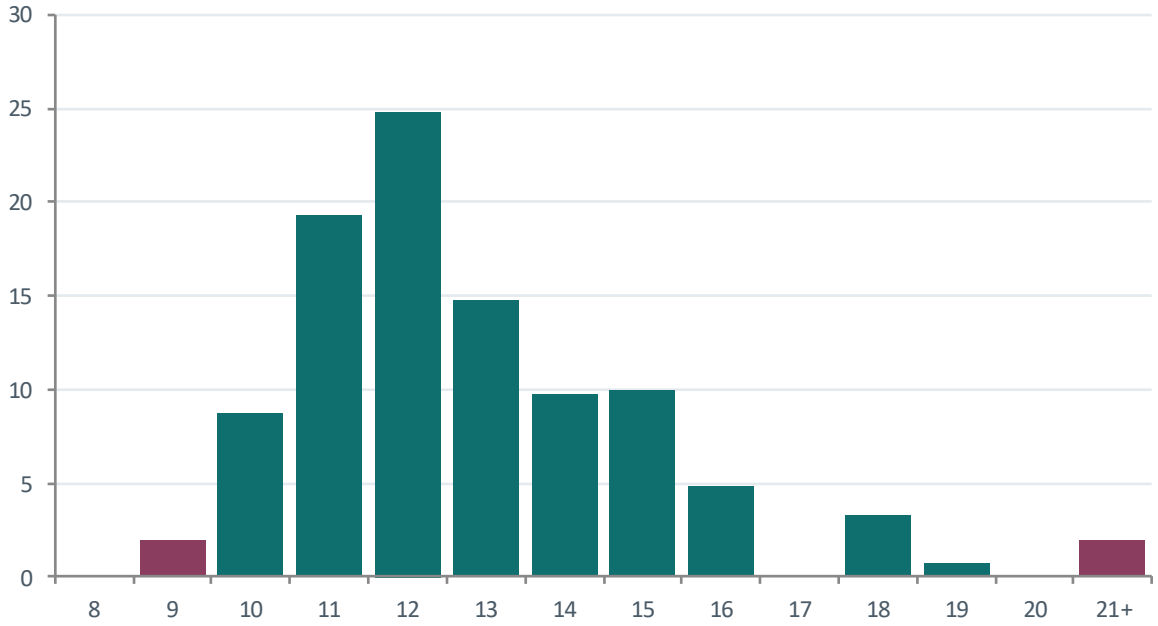
Original	7	9	10	11	12	13
	15	18	24	36		

13 days

Redrawn	9	9	10	12	13	13
	15	24	36	36		

9 and 13 and 36 drawn twice; 7, 11, 18 not at all. Median of this redraw: 13 days.

Ten thousand medians



Median wait from each of 10 000 redrawn samples (% of samples)

The percentile interval

Sort the 10 000 medians. Cut off the lowest 2.5% and the highest 2.5%. What remains is the interval.

12.5	95% CI	10.0	19.5
median wait (days)		lower	upper

The whole thing, in R

```
wait <- c(12, 18, 9, 36, 13, 24, 15, 11, 7, 10)   # days

meds <- replicate(10000, median(sample(wait, 10, replace = TRUE)))

median(wait)
quantile(meds, c(0.025, 0.975))      # 95% confidence interval
```

Swap median() for any function and the code still works.

Try it with mean, sd, or quantile(wait, 0.9)

One keyword changes everything

```
sample(12, 6)                    # permutation: deal the cards, none repeat
sample(x, 10, replace = TRUE)    # bootstrap: put each card back after drawing
```

replace = FALSE

Rearranges what you have. Answers: could chance alone have produced this difference?

replace = TRUE

Imitates drawing new patients. Answers: how much would the answer move in a repeat study?

CLUSTERED DATA

Several teeth per patient

The same twelve measurements — but they came from six patients, two teeth each. The treatment was given to the patient, not to the tooth.

P1	P2	P3	P4	P5	P6
3.1 2.8	4.0 3.6	2.9 3.4	2.4 2.1	3.0 2.6	2.2 2.8

Treatment A: patients 1-3. Treatment B: patients 4-6.

Twelve teeth, but only six independent pieces of information

Clustered data

Two ways to handle it

1 · Collapse

Average the teeth in each mouth. Twelve teeth become six patient values — and from there, every method from parts 1 and 2 works unchanged.

Simplest. Use this by default.

2 · Resample whole patients

Keep every tooth, but draw or shuffle at the level of the patient, so a mouth always travels intact.

Needed when patients contribute different numbers of teeth.

3

Sample size from a pilot study

The most useful thing a small study can tell you is how big the next one must be.

SAMPLE SIZE

What a pilot study is actually for

Not for

Testing the hypothesis. Its p-value is noise, and its effect size is noise dressed as a result.

But for

Measuring how much your outcome scatters — the standard deviation — so the real study can be sized.

```
pilot <- c(1.1, 0.4, 1.6, 0.9, 1.3, 0.2, 1.5, 0.8)    # 8 patients
sd(pilot)                                              # 0.50
power.t.test(delta = 0.5, sd = sd(pilot), power = 0.8)$n
```

The answer, and the catch

17

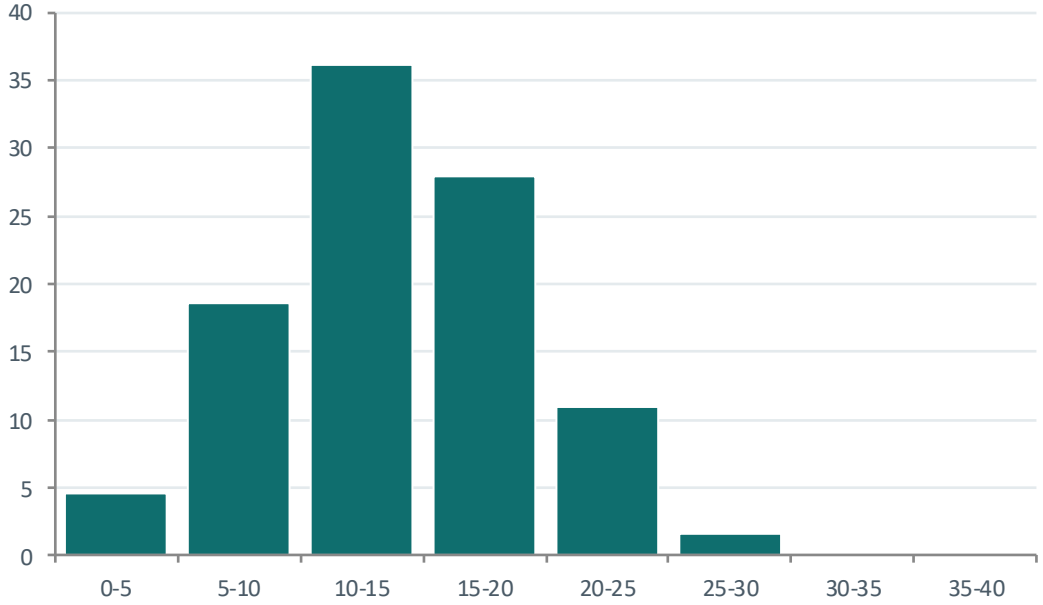
patients per group, from the pilot's SD of 0.50

The probem

That 0.50 came from eight patients. It is an estimate too — and a shaky one.

```
ns <- replicate(2000, {
  s <- sd(sample(pilot, 8, replace = TRUE))
  power.t.test(delta = 0.5, sd = s, power = 0.8)$n
})
quantile(ns, c(0.025, 0.975))
```

Plan for the upper end (improve your ss calculation)



Required n per group across 2 000 bootstrapped pilots (% of pilots)

5 – 25

plausible range for n per group

So what?

Budget for the upper end

Four things to take away

1. **How many replicates?**
   At least 2 000 for a p-value, 10 000 for a confidence interval. It costs seconds.

2. **Resample the independent unit**
   The patient — not the tooth, not the surface, not the visit.

3. **Permutation and bootstrap answer different questions**
   Is there a difference? versus how big is it, and how sure are we?

4. **Resampling relaxes assumptions only**
   It cannot repair bias, confounding, or a study that was too small.